AZZUNIQE CRM & ERP SOFTWARE

MANUFACTURING & PRODUCTION SERVICE

Detailed Super Admin Functional Documentation

Formula / BOM + Production | No Database Table/Field Listing

Scope: This document extracts only the Manufacturing & Production service from the main Super Admin documentation. Database table names and field-by-field definitions are intentionally excluded. All functional flow, sub-modules, frontend behavior, APIs, permissions, approvals, audit/notification behavior and service integration are retained.

1. Service Overview

Service: Manufacturing & Production

Main Modules: Formula / BOM and Production

Core purpose: Product -> Formula/BOM -> Version -> Approval -> Active Formula -> Production, followed by Sales/Stock Need -> Production Order -> Material Check -> Reserve/Issue -> Batch -> DPR -> Consumption/Wastage -> QC -> Finished Goods -> Inventory.

2. High-Level End-to-End Flow

- Product requirement is linked to a Formula / BOM.
- Formula / BOM is prepared as a controlled version with materials, quantities and instructions.
- Formula version is submitted for approval.
- Only an approved active formula is used by Production.
- A Sales/Stock Need creates the requirement for a Production Order.
- Production checks material availability and reserves/issues raw material from inventory.
- Production is executed against a batch/lot.
- Daily Production Reports (DPR) record production progress and output.
- Actual material consumption and wastage/scrap are recorded.
- QC records pass, fail or rework result for the production batch.
- Accepted finished quantity is posted as Finished Goods into Inventory.

3. Module 1 - Formula / BOM

Module workflow: Product -> Formula/BOM -> Version -> Approval -> Active Formula -> Production

3.1 Formula / BOM Sub-Modules and Operations

Sub-Module / Screen	Detailed Functional Behavior
Formula / BOM	Create and edit a draft formula; manage version, production instructions and yield; submit for approval; approve/reject where authorized; activate the approved version.

Formula Items	Add, edit or remove raw materials and their required quantities while the formula version is still in an editable draft state.
Instructions	Maintain the production/process instructions and notes associated with the formula.
Versions	Create a new version while preserving prior approved versions so production history is not silently overwritten.
Approval	Submit, approve or reject a formula version. Production must use the approved active version.

3.2 Formula / BOM Frontend Page Flow

Step	Business / Frontend Action	System Behavior
1	Open Module	Show Formula / BOM dashboard/list and only those Sub-Modules available under the user's VIEW permission.
2	Open Sub-Module	Load list data with search, filter, sort and pagination. Show +Add only when CREATE permission is available.
3	Create / Edit	Open the form, load dependent relation selections, validate entries, save through create/update operation, then refresh list/detail.
4	Detail / Children	Open formula detail with related child rows/tabs and show only actions allowed by permission and current status.
5	Workflow Actions	Submit, approve, reject and activate through dedicated workflow actions where applicable.
6	Audit / Notification	Important create, update, status and approval events write audit/status history and notify the next responsible user when configured.

3.3 Formula / BOM APIs

Method	Endpoint	Purpose
GET / POST / PUT / DELETE	/api/formulas	Formula list, create, detail/update and delete operations.
GET / POST	/api/formulas/:id/items	Read/add Formula Items under a formula.
POST	/api/formulas/:id/submit	Submit a formula version for approval.
POST	/api/formulas/:id/approve	Approve the submitted formula version.
POST	/api/formulas/:id/activate	Make the approved formula version active for Production.

3.4 Formula / BOM Permission Model

- Scope: Manufacturing & Production -> Formula / BOM -> each Sub-Module.
- Actions: VIEW, CREATE, EDIT, DELETE, SUBMIT, APPROVE, REJECT.
- Permission code pattern: <service_code>.<module_code>.<sub_module_code>.<action>.
- Effective access requires an active permission, active role/user assignment and valid owner scope; a user-specific override wins over the role default.

4. Module 2 - Production

Module workflow: Sales/Stock Need -> Production Order -> Material Check -> Reserve/Issue -> Batch -> DPR -> Consumption/Wastage -> QC -> Finished Goods -> Inventory

4.1 Production Sub-Modules and Operations

Sub-Module / Screen	Detailed Functional Behavior
Production Order	Create and plan a production order; select product, approved formula, quantity and date; release/start production; complete or cancel according to status.
Batch	Create a production batch/lot and track its status and produced quantity throughout production.
Material Issue	Reserve and then issue raw material. Validate available batch stock before material is issued to production.
DPR	Create/edit Daily Production Report entries including production output, shift and remarks.
Consumption	Record actual material consumption against the production batch/order and retain visibility of planned versus actual usage through the production flow.
Wastage / Scrap	Record wastage or scrap type, quantity and reason; approval is required where the configured threshold is exceeded.
QC	Create quality inspection and record PASS / FAIL / REWORK with inspection remarks/results.
Finished Goods	Post accepted finished quantity into the inventory batch after successful production/QC processing.
Contract Manufacturing	Track outsourced production reference, progress and receipt where contract manufacturing is configured.

4.2 Production Frontend Page Flow

Step	Business / Frontend Action	System Behavior
1	Open Module	Show Production dashboard/list and visible Sub-Modules according to VIEW permission.
2	Open Sub-Module	Load list with search/filter/sort/pagination and show +Add only when CREATE is permitted.
3	Create / Edit	Open form, load related selections, validate, save through create/update operation and refresh the current list/detail.
4	Detail / Children	Open a production record with child rows/tabs and display actions according to permission and current workflow status.
5	Workflow Actions	Use dedicated actions for issue, complete and other production workflow transitions where applicable.
6	Audit / Notification	Important create/update/status/workflow actions write audit/status history and notify the next responsible user when configured.

4.3 Production APIs

Method	Endpoint	Purpose
GET / POST / PUT / DELETE	/api/production-orders	Production Order list/create/update/delete and detail operations.
GET / POST	/api/production-orders/:id/batches	Production batch/lot operations.
GET / POST	/api/production-orders/:id/material-issues	Reserve/issue material records for the Production Order.
GET / POST	/api/production-orders/:id/dprs	Daily Production Report operations.
GET / POST	/api/production-orders/:id/ consumption	Record/read material consumption.
GET / POST	/api/production-orders/:id/wastage	Record/read wastage and scrap.
GET / POST	/api/production-orders/:id/qc	Record/read QC inspection.
GET / POST	/api/production-orders/:id/finished-goods	Post/read accepted finished goods.

4.4 Production Permission Model

- Scope: Manufacturing & Production -> Production -> each Sub-Module.
- Actions: VIEW, CREATE, EDIT, DELETE, COMPLETE, ISSUE.
- Permission code pattern: <service_code>.<module_code>.<sub_module_code>.<action>.
- Effective access requires active permission, active role/user assignment and valid owner scope; user-specific override wins over role default.

5. Detailed Business Flow Between Formula/BOM and Production

- Formula / BOM is the controlled definition used by Production. A formula must reach the approved active state before it is used in the production process.
- Production Order is created from a Sales/Stock Need and references the product and its approved formula.
- Material Check happens before issue. Material is reserved/issued only from available inventory batch stock.
- Production execution is tracked through a production batch/lot.
- DPR records daily production progress/output while Consumption captures actual material usage.
- Wastage/Scrap is recorded separately with type, quantity and reason, and approval applies where a configured threshold is exceeded.
- QC is the decision point for PASS, FAIL or REWORK.
- Only accepted finished quantity is posted into finished-goods inventory.
- For outsourced work, Contract Manufacturing tracks the external production reference, progress and receipt where configured.

6. Status, Approval, Audit and Notification Controls

- Frontend action buttons are permission-aware and also depend on the current record status.
- Formula workflow supports submit, approve, reject and activate behavior.
- Production workflow supports controlled release/start, material issue, QC and completion behavior.
- Important create/update/status/approval actions write audit/status history.
- Configured notifications are sent to the next responsible user after important workflow events.
- The frontend hiding of actions is not the security boundary; backend APIs must enforce ownership, active scope and effective action permission.

7. Functional Integration Points

Connected Area	Manufacturing & Production Relationship
Product	Formula/BOM belongs to a finished product and Production Order is created for the product.
Sales / Stock Need	Demand is the trigger for Production Order creation.
Inventory & Warehouse	Material availability is checked before reserve/issue; issued material is taken from inventory batches.
Quality Control	QC determines PASS, FAIL or REWORK for the production batch.
Finished Goods / Inventory	Accepted finished quantity is posted back into inventory after production and QC.

8. Service Completion Summary

Internal service order: Formula / BOM -> Production.

The Manufacturing & Production service is complete when the controlled formula/version process, production execution, material issue, DPR, consumption, wastage/scrap, QC and finished-goods posting are operating together under the configured permissions, workflow controls, audit history and notifications.